



# Instructor Solution.

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

Time = 15 Minutes

## Instructor Solution - Quiz No 3

[20 points]

Note - Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

### Question 1 [08 points] Counting

From 5 seniors, 4 juniors, 1 sophomore and 8 freshmen, in how many ways can the group of five be chosen if there must be at least one member from each class?

### Question 2 [12 points] Calculating Probability using Counting

Given that there were 4 heads in 9 tosses, what is the probability that the first three tosses were heads? Assume it is a biased coin with  $P(H) = p$ .

You are required to solve this using **ONLY** the counting method and using the following two events. You need to back your working with solid reasoning. This should definitely include the reasoning, how can the required probability be calculated using the counting methods.

- event A: the first 3 tosses were heads.
- event B: 4 out of 9 tosses were heads.

$$\text{Note - } P(A | B) = \frac{P(A \cap B)}{P(B)} =$$

Q1 - There are 5 ways to select a senior, 4 ways to select a junior, one way to select a sophomore and 8 ways to select a freshmen. (2)

8 pts  $\Rightarrow 5C_1 \cdot 4C_1 \cdot 1C_1 \cdot 8C_1 = 5 \times 4 \times 1 \times 8 = 160 \text{ ways}$

Therefore, in 160 ways we can select a subset with 4 elements containing 1 member of each class. (1)

(1) Remaining Students =  $18 - 4 = 14$  members left

$\Rightarrow$  we may choose any one of the 14 students to complete the group of 5.

$$\Rightarrow 160 \times \frac{14}{2} = 1120 \text{ — } \textcolor{red}{(3)}$$

we divide by 2 because each set of 5 elements occurs twice. (1)

Q2) Since it is a biased Coin, Each particular Sequence <sup>(w/ 9 tosses)</sup> has a different probability given by :-

$$P(\text{particular Seq with } K\text{-Heads}) = p^K (1-p)^{9-K}$$

However, given there are exactly 4 Heads in 9 tosses i.e. Event B, all such Elements in B have equal probability given below

$$\textcircled{1} P(\text{Sequence w/ 4-Heads}) = p^4 (1-p)^5 \text{ --- } \textcircled{A_1}$$

Given B, all outcomes have probability as an  $\textcircled{A_1}$  (not related to Event A)

Within this Set B, are Sequences that

$\textcircled{3}$  has first 3 tosses as Heads i.e. Event A.

So probability,  $P(A|B)$  can be calculated using No of ~~Seq~~ Elements in  $(A \cap B)$  normalized by Number of Elements in B.

$$\text{No of Elements in } B = {}^9C_4 = \textcircled{2}$$

$$\text{No of Elements in } (A \cap B) = (H_1, H_2, H_3) \text{ One in 6 is a Head (only)}$$

$$\textcircled{4} = {}^6C_1 = 6 \text{ --- } \textcircled{2}$$

$$\text{therefore, } P(A|B) = \frac{6}{\textcircled{2}} = \frac{6}{126} = 0.0476$$